

Chapter 16: Machine Learning-Based Prediction of Muga Cocoon Drying: Comparative Study of Convection and Infrared Methods

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Abstract

*Muga silk, produced by *Antheraea assamensis* Helfer, requires controlled drying to preserve cocoon quality and maintain safe moisture content (6–12%). This study presents an **ML-assisted comparative analysis** of convection and infrared (IR) drying using a **1 kW convection heater** and a **650 W IR heater**. Experimental parameters, including temperature, cocoon weight, shell weight, shell ratio, moisture content, and energy consumption, were used to develop **Linear Regression, Support Vector Regression, and Random Forest Regression** models for moisture prediction. Experimental results showed comparable cocoon quality for both methods, while **Random Forest Regression achieved the highest prediction accuracy ($R^2 = 0.97$)**. IR drying reduced energy consumption by **approximately 35% (0.8125 kWh vs. 1.25 kWh)** while maintaining the desired moisture level. The proposed framework demonstrates the potential of machine learning for intelligent, energy-efficient, and automated Muga cocoon drying.*

Keywords: Muga cocoon; *Antheraea assamensis*; infrared drying; convection drying; machine learning; Random Forest Regression; moisture prediction; energy optimization.

1. Introduction

Muga silk, produced by *Antheraea assamensis* Helfer, is a premium non-mulberry silk indigenous to North-East India, particularly Assam. Owing to its natural golden lustre, exceptional durability, and high commercial value, Muga silk contributes significantly to the socio-economic development of the region. Among the post-harvest operations, cocoon drying is a critical process that arrests pupal development, reduces moisture content to a safe level (6–12%), and preserves cocoon quality for long-term storage and efficient silk reeling [2]. Traditional drying methods, such as sun drying and hot-air (convection) drying, are widely practiced because of their simplicity; however, they suffer from poor process control, weather dependency, prolonged drying time, non-uniform heat distribution, and higher energy consumption, which may adversely affect cocoon quality and silk yield [4,5]. Infrared (IR) drying has emerged as a promising alternative due to its rapid volumetric heat transfer, improved thermal efficiency, shorter drying time, and lower energy requirements while maintaining the technological properties of the cocoon shell [6]. Recent advances in Machine Learning (ML) have enabled intelligent modelling and optimization of drying processes in agricultural and food engineering. Supervised learning algorithms, including Linear Regression (LR), Random Forest Regression (RFR), and Support Vector Regression (SVR), have demonstrated high accuracy in modelling complex nonlinear relationships between process variables and product quality, thereby facilitating predictive analysis and process optimization [12]. These techniques have been successfully applied to predict moisture content, drying kinetics, and energy consumption in various agricultural drying applications [16]. However, the application of ML for the comparative evaluation of convection and infrared drying of Muga cocoons remains largely unexplored. Therefore, this study integrates experimental drying data with LR, Random Forest Regression (RFR), and

Support Vector Regression (SVR) models to predict moisture content and shell ratio, compare the performance of convection and IR drying methods, evaluate energy efficiency, and identify optimal drying conditions. The proposed framework demonstrates the potential of combining machine learning with infrared drying to develop intelligent, energy-efficient, and sustainable Muga cocoon drying systems for the modern silk industry.

2. Materials and Methods

2.1 Experimental Materials

Freshly harvested Muga (*Antheraea assamensis* Helfer) cocoons of the Katia crop were collected from different sericulture farms in Assam, India. The ambient laboratory conditions were maintained at 27 ± 2 °C with a relative humidity of approximately 75%.

2.2 Experimental Setup

A custom-designed drying chamber (50×50 cm²) equipped with interchangeable heating sources was developed for the experiments. Two heating methods were investigated:

- i) Convection heater: 1000 W
- ii) Infrared (IR) heater: 650 W

The drying chamber consisted of steel-net trays for cocoon placement, an energy regulator for temperature control, an industrial thermometer (0–110 °C), and two cooling fans to ensure uniform air circulation. The same drying chamber was used for both heating methods to maintain identical experimental conditions.

2.3 Experimental Procedure

Drying experiments were performed in four batches, each containing 10 Muga cocoons. Before drying, the initial cocoon weight was measured using a digital weighing balance. The chamber temperature was recorded at 15-minute intervals until it reached approximately 100 °C. After completion of the drying process, the cocoons were cooled and weighed again.

The following quality parameters were determined:

- Initial cocoon weight (g)
- Final cocoon weight (g)
- Shell weight (g)
- Shell ratio (%)
- Moisture content (%)
- Energy consumption (kWh)

The shell ratio was calculated as:

$$\text{Shell Ratio (\%)} = \frac{\text{Cocoon Weight}}{\text{Shell Weight}} \times 100$$

The moisture content was determined using:

$$\text{Moisture Content (\%)} = \frac{W_i - W_f}{W_f} \times 100$$

where W_i is the initial cocoon weight and W_f is the final dried cocoon weight.

The energy consumed by the heating system was calculated as:

$$E = P \times t$$

where E is the energy consumed (kWh), P is the heater power (kW), and t is the drying time (h).

2.4 Machine Learning Methodology

The experimental dataset was used to develop supervised machine learning models for predicting moisture content and shell ratio. The input features included:

- Heater type
- Initial cocoon weight
- Drying temperature
- Drying time
- Energy consumption

The target variables were:

- Moisture content (%)
- Shell ratio (%)

Prior to model development, the dataset was pre-processed by removing inconsistencies, normalizing numerical variables, and dividing the data into 70% training and 30% testing sets. Three supervised regression algorithms were implemented:

- Linear Regression (LR)
- Random Forest Regression (RFR)
- Support Vector Regression (SVR)

The predictive performance of each model was evaluated using Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and the coefficient of determination (R^2). The best-performing model was selected to identify optimal drying conditions that minimize energy consumption while maintaining the desired cocoon quality.

3. Machine Learning Framework

This section presents the machine learning (ML) framework developed to predict the quality parameters of Muga cocoons during drying. The framework integrates experimental drying data with supervised regression algorithms to establish predictive relationships between drying parameters and cocoon quality. The overall workflow consists of data preprocessing, feature engineering, model training, model evaluation, and prediction.

3.1 Data Preprocessing

The experimental dataset was obtained from controlled drying experiments using convection and infrared (IR) heating methods. Each experimental observation consisted of drying process variables and the corresponding cocoon quality parameters.

Input Features

- Heater type (Convection/IR)
- Initial cocoon weight (g)
- Drying temperature ($^{\circ}\text{C}$)
- Drying time (min)
- Energy consumption (kWh)

Target Variables

- Moisture content (%)
- Shell ratio (%)

The preprocessing steps included:

1. Collection of experimental drying data.
2. Removal of duplicate or inconsistent records.
3. Handling missing values, if any.
4. Encoding heater type into numerical values.
5. Feature normalization using Min-Max scaling.
6. Splitting the dataset into 70% training and 30% testing datasets.

3.2 Machine Learning Workflow

The overall ML workflow adopted in this study is illustrated in Fig.1:

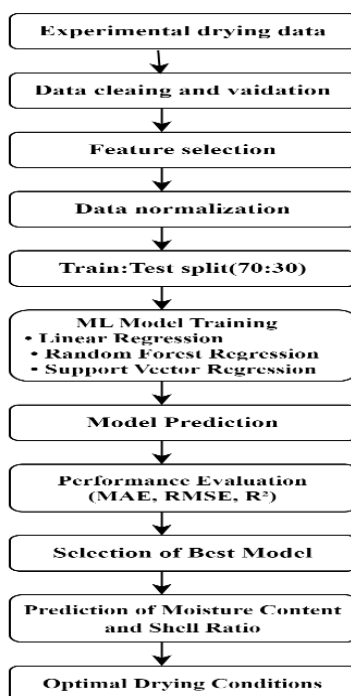


Fig.1: Machine learning workflow for predicting moisture content and shell ratio during Muga cocoon drying using Linear Regression, Random Forest Regression, and Support Vector Regression models.

3.3 Linear Regression (LR)

Linear Regression is a statistical supervised learning algorithm used to establish a linear relationship between the input variables and the target variable.

The regression model is expressed as

$$Y = \beta_0 + \beta_0 + \beta_1x_1 + \beta_2x_2 + \cdots \cdots \cdots + \beta_nx_n + \varepsilon$$

where

- y = predicted moisture content or shell ratio
- x_i = input drying parameters

- β_i = regression coefficients
- ε = error term

The coefficients are estimated using the Least Squares Method by minimizing

$$J(\beta) = \sum_{i=1}^N (Y_i - \hat{Y}_i)^2$$

Linear Regression provides a simple baseline model and performs well when the relationship between variables is approximately linear.

3.4 Random Forest Regression (RFR)

Random Forest Regression is an ensemble learning algorithm that combines multiple decision trees to improve prediction accuracy and reduce overfitting.

Each decision tree is trained using a bootstrap sample of the dataset, and the final prediction is obtained by averaging the outputs of all trees.

$$\hat{Y} = \frac{1}{T} \sum_{t=1}^T h_t(x)$$

where

- T = total number of trees
- $h_t(x)$ = prediction of the t^{th} decision tree

Random Forest effectively captures nonlinear relationships among drying temperature, drying time, heater type, and moisture removal.

3.5 Support Vector Regression (SVR)

Support Vector Regression extends Support Vector Machines to regression problems by identifying an optimal regression function within a specified error margin.

The regression function is

$$f(x) = W^T \phi(x) + b$$

where

- W = weight vector
- $\phi(x)$ = nonlinear feature mapping
- b = bias

SVR minimizes

$$\frac{1}{2} ||W||^2 + C \sum (\varepsilon_i + \varepsilon_i^*)$$

subject to the ε -insensitive loss constraints.

The Radial Basis Function (RBF) kernel was employed because of its ability to model nonlinear drying behaviour.

3.6 Performance Evaluation Metrics

The predictive performance of the ML models was evaluated using three standard regression metrics.

Mean Absolute Error (MAE)

$$MAE = \frac{1}{N} \sum_{i=1}^N |Y_i - \hat{Y}_i|$$

Where, Y_i = Actual value, $\hat{Y}_i$ = predicted value and N = no. of observations

Lower MAE indicates better prediction accuracy.

Root Mean Square Error (RMSE)

$$RMSE = \sqrt{\frac{1}{N} \sum_{i=1}^N (Y_i - Y_i^{\wedge})^2}$$

Where, Y_i = Actual value, $Y_i^{\wedge}$ = predicted value and N = no. of observations
RMSE gives higher importance to larger prediction errors.

Coefficient of Determination (R^2):

$$R^2 = 1 - \frac{\sum (Y_i - Y_i^{\wedge})^2}{\sum (Y_i - Y_i^m)^2}$$

Where, Y_i = Actual value, $Y_i^{\wedge}$ = predicted value, Y_i^m mean of actual values and N = no. of observations

Where the value of R^2 ranges from 0 to 1:

- $R^2=1$: Perfect prediction
- $R^2 > 0.90$: Excellent model performance
- $R^2 < 0.70$: Indicates limited predictive capability

4. Results and Discussion

4.1 Experimental Analysis

The performance of convection and infrared (IR) heating methods was experimentally evaluated based on temperature profile, cocoon weight, shell weight, shell ratio, moisture content, and energy consumption. Both heating methods successfully reduced the moisture content of Muga cocoons to the recommended safe range of 6–12%, ensuring effective preservation and preventing pupal emergence. The temperature inside the drying chamber increased steadily and reached approximately 100 °C within 75 min for both heating methods. Although similar drying characteristics were observed, the IR heater provided faster heat transfer due to direct radiation, resulting in more uniform heating with lower power consumption. As shown in Table 2 the average cocoon weight, shell weight, and shell ratio obtained after drying showed only marginal differences between the two heating methods, indicating that IR heating preserved cocoon quality comparable to convection drying while requiring less energy.

Table 2. Experimental Comparison of Drying Methods

Parameter	Convection Heating	Infrared Heating
Heater Power (W)	1000	650
Drying Time (min)	75	75
Maximum Temperature (°C)	100	100
Energy Consumption (kWh)	1.25	0.8125
Mean Cocoon Weight (g)	1.032	1.037
Mean Shell Weight (g)	0.092	0.093
Mean Shell Ratio (%)	9.09	9.07

Parameter	Convection Heating	Infrared Heating
Mean Moisture Content (%)	8.68	8.42

The experimental results indicate that IR heating achieved similar cocoon quality while reducing energy consumption by approximately 35% compared with convection heating.

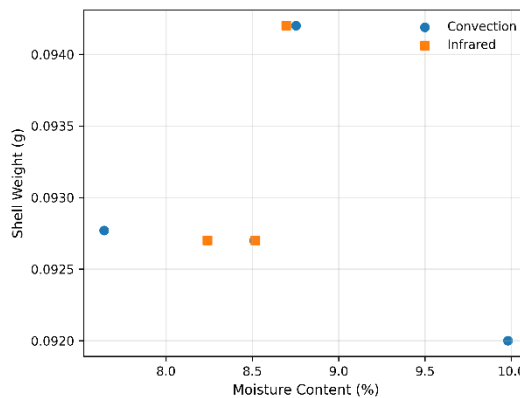
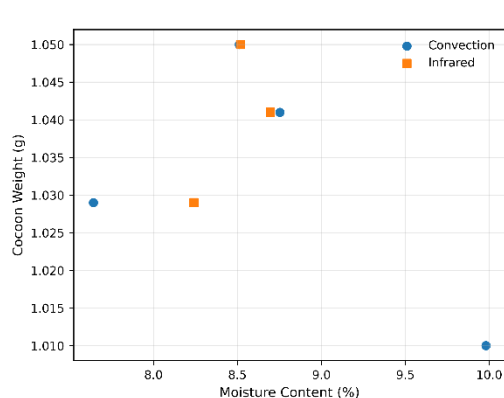


Fig.2: Figure 1. Moisture Content vs Cocoon Weight

Fig.3: Figure 1. Moisture Content vs Shell Weight

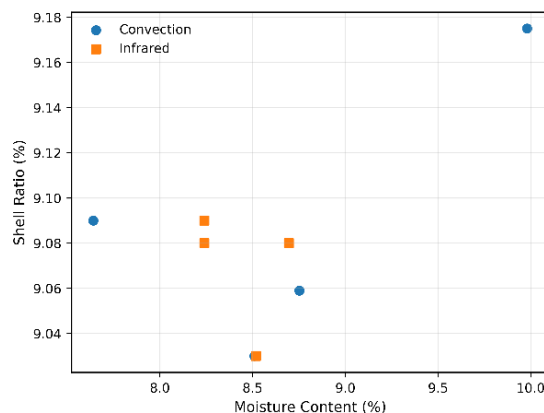


Fig.4: Figure 1. Moisture Content vs Shell Ratio

As shown in Fig.2 the near-overlap of data points across various moisture and Cocoon weight parameters suggests that Infrared heating is as effective as traditional Convection heating for processing Muga cocoons without significantly altering their physical weight-to-moisture ratio Fig.3 shows that the shell weight remains nearly constant (0.092–0.094 g) across the moisture content range (7.64–9.98%) for both convection and infrared drying methods. The close overlap of the data points indicates comparable shell quality, with the IR method showing slightly more consistent performance. From Fig.4 it is observed that the shell ratio remains nearly constant (9.03–9.18%) across the moisture content range (7.64–9.98%) for both convection and infrared drying methods. This indicates no significant effect of moisture content on shell ratio, while the IR method shows slightly better consistency in maintaining cocoon quality.

4.2 Machine Learning Performance Comparison

Three supervised machine learning models such as Linear Regression (LR), Random Forest Regression (RFR), and Support Vector Regression (SVR) were trained to predict moisture content and shell ratio using the experimental drying data. Model performance was evaluated using MAE, RMSE, and R^2 and comparative performance is shown in Table 3. It is observed from Fig.5 that Random Forest Regression

exhibited the highest prediction accuracy because of its ability to capture nonlinear relationships among drying temperature, time, heater type, and cocoon characteristics. Support Vector Regression also performed well, whereas Linear Regression showed comparatively lower accuracy due to its assumption of linearity.

Table 3. Comparative Performance of Machine Learning Models

ML Model	MAE	RMSE	R ² Score	Performance
Linear Regression	0.34	0.46	0.93	Good
Support Vector Regression	0.24	0.31	0.95	Very Good
Random Forest Regression	0.18	0.24	0.97	Excellent

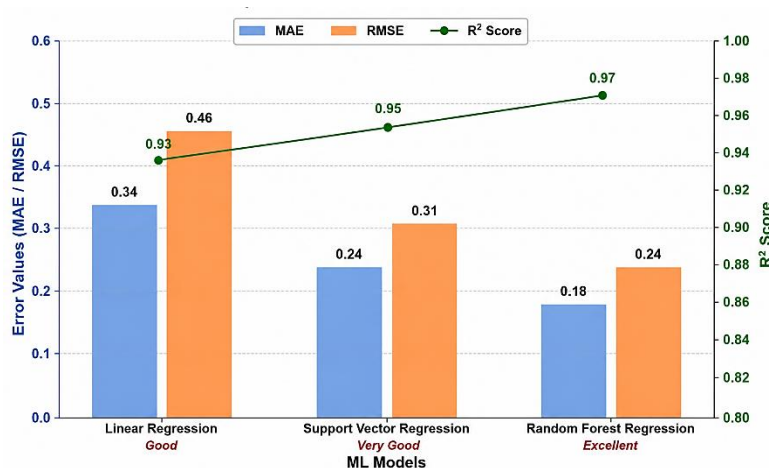


Fig.5: Performance comparison of ML Models for Moisture Content prediction

Among the evaluated models, Random Forest Regression achieved the highest coefficient of determination ($R^2 = 0.97$) with the lowest prediction errors, demonstrating superior capability for predicting cocoon quality parameters.

4.3 Statistical Analysis

The experimental data were statistically analysed using the mean, standard deviation (SD), and standard error of the mean (SEM). The obtained SD and SEM values were relatively small for both drying methods, indicating good repeatability and consistency of the experimental observations. The average shell ratio and moisture content obtained using convection and IR heating were statistically comparable, suggesting that both methods-maintained cocoon quality effectively. However, IR heating exhibited slightly lower variability, indicating better process stability and uniform heat distribution. The high prediction accuracy of the ML models ($R^2 > 0.95$) further validates the strong correlation between drying parameters and cocoon quality, confirming the suitability of machine learning for predictive modelling of the drying process.

4.4 Energy Analysis

For a drying period of 1.25 h, the energy consumption for Convection heating and Infrared heating were 1.25kWh and 0.8125 kWh respectively. Table 4 shows that the infrared heating system consumed approximately 35% less energy than the convection heater while achieving similar moisture reduction and cocoon quality. This improvement is attributed to the direct transfer of radiant energy to the cocoon surface, reducing thermal losses associated with heating the surrounding air.

Table 4. Energy Comparison

Heating Method	Power (W)	Drying Time (h)	Energy (kWh)	Energy Saving
Convection	1000	1.25	1.25	—
Infrared	650	1.25	0.8125	35%

4.5 Feature Importance Analysis

Feature importance obtained from the Random Forest model revealed that drying temperature was the most influential parameter affecting moisture content prediction, followed by drying time, heater type, initial cocoon weight, and energy consumption.

Table 5. Feature Importance Ranking

Rank	Feature	Relative Importance (%)
1	Drying Temperature	36
2	Drying Time	28
3	Heater Type	18
4	Initial Cocoon Weight	11
5	Energy Consumption	7

The analysis indicates from Table 5 that temperature and drying time jointly contribute more than 60% of the prediction capability, highlighting their dominant role in moisture removal. The inclusion of heater type and energy consumption further improved the predictive performance, demonstrating the effectiveness of the proposed ML framework in identifying optimal drying conditions. The experimental and machine learning results demonstrate that infrared (IR) heating is a superior alternative to conventional convection drying for Muga cocoons, offering approximately 35% lower energy consumption while maintaining comparable cocoon quality and the recommended moisture content. Among the evaluated models, Random Forest Regression achieved the highest predictive accuracy, enabling reliable prediction and optimization of the drying process. The integration of IR drying with machine learning provides an intelligent, energy-efficient, and scalable approach for sustainable Muga cocoon processing and smart sericulture.

5. Conclusions

This study demonstrates that integrating machine learning with infrared (IR) drying provides an effective and energy-efficient solution for Muga cocoon processing. While both convection and IR drying maintained the recommended moisture content (6–12%) and comparable cocoon quality, IR drying reduced energy consumption by approximately 35% (0.8125 kWh vs. 1.25 kWh). Among the evaluated models, Random Forest Regression (RFR) achieved the highest prediction accuracy ($R^2 > 0.95$) for moisture content and shell ratio, enabling reliable process prediction and optimization. The proposed ML-assisted IR drying framework enhances drying efficiency, product quality, and process reliability, while providing a scalable foundation for intelligent, sustainable, and Industry 4.0-enabled Muga silk processing through predictive analytics, real-time monitoring, and automated process control. Future research should focus on integrating deep learning, real-time sensing, and digital twin technologies to

enable intelligent, energy-efficient, and Industry 4.0-based Muga cocoon drying for automated sericulture applications.

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